Consensus Control for Multi-Agent Systems under Switching Topologies via State-Dependent Dwell-Time Switching

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***Abstract*—This paper investigates the consensus problem for switched multi-agent systems with multiple candidate directed communication topologies. Each candidate topology is allowed to be disconnected, while the union graph of all candidate topologies is assumed to be strongly connected. A consensus-orthogonal decomposition is first introduced to remove the consensus direction and transform the original multi-agent system into a reduced-order switched disagreement error system. Based on this formulation, the strong connectivity of the union graph is used to exclude common non-consensus zero directions among the candidate topologies. A state-dependent dwell-time switching law is then designed via Lyapunov–Metzler inequalities to stabilize the disagreement error system. Under the jointly strongly connected condition, a strictly positive convex weight vector and a sufficiently large coupling parameter can be selected such that the associated zero-dwell-time lifted matrix is Hurwitz. Consequently, the Lyapunov–Metzler inequalities are feasible for every prescribed dwell time below an explicitly derived positive upper bound. It is shown that, if the Lyapunov–Metzler inequalities are feasible, the disagreement error converges to zero and global consensus is achieved. A numerical example with two individually disconnected topologies is provided to illustrate the effectiveness of the proposed approach.**

***Index Terms*—Multi-agent consensus, switching topologies, jointly strongly connected, Lyapunov–Metzler inequalities, state-dependent dwell-time.**

I. Introduction

Consensus control is a fundamental problem in multi-agent systems, where a group of agents is required to reach agreement via local information exchange over communication networks [1], [2]. Owing to its broad applications in cooperative control, distributed sensing, formation control, unmanned systems, and networked autonomous systems, consensus has been extensively studied. In graph-based consensus analysis, connectivity of the communication topology is the standard condition for undirected networks [3]. For directed networks, typical graph-theoretic conditions require the communication graph to be strongly connected or to contain a directed spanning tree [4], [5]. These results reveal that sufficient information propagation across the network is essential for achieving consensus.

In practical networked systems, however,the communication topology may switch over time due to link failures, sensing limitations, communication constraints, cyber attacks, or intermittent information exchange. Under such switching topologies, each instantaneous graph may be disconnected and may fail to guarantee global consensus by itself. Nevertheless,the collection of switching graphs may jointly provide sufficient information paths over time. This motivates the study of consensus under switching topologies with joint or union graph connectivity. Existing results have shown that consensus can still be achieved when the switching graphs satisfy jointly connected conditions, even though each instantaneous graph may not be connected [6]–[8].

Since switching topologies lead to a switched consensus dynamics, an effective way to analyze convergence is to separate the consensus component from the disagreement component. By using an orthogonal transformation, the original consensus problem can be converted into the stability problem of a reduced-order disagreement error system [9], [10]. The resulting model has no redundant consensus direction and can therefore be analyzed directly by switched-system stability tools.

The stability analysis of switched systems with possibly unstable subsystems provides another relevant perspective for studying switching-induced consensus dynamics. Classical switched-system theory has shown that a suitable switching law may stabilize the overall system even when some subsystem matrices are unstable or marginally stable [11], [12], [14]. More recent studies have developed mode-dependent dwell-time and multiple-Lyapunov-function methods for switched systems containing unstable modes [16]–[19].

Recently, Russo et al. proposed a state-dependent dwell-time switching framework for switched affine systems based on differential Lyapunov equations and Lyapunov–Metzler inequalities [19]. Their general framework demonstrates that stabilization under a dwell-time constraint does not require all subsystem matrices to be Hurwitz. Inspired by this idea, this paper applies a Lyapunov–Metzler-based dwell-time switching mechanism to the reduced-order disagreement dynamics of switched multi-agent systems with multiple candidate directed communication topologies. Each candidate topology is disconnected, whereas the union graph of all candidate topologies is assumed to be strongly connected. After removing the consensus direction through an orthogonal decomposition, the original consensus problem is reformulated as a stabilization problem for a reduced-order switched disagreement system. The resulting switching law enforces a prescribed minimum dwell time and selects the active communication topology according to the current disagreement state. Under the union-connectivity condition, an Metzler matrix is constructed, and the feasibility of the finite-dwell-time Lyapunov–Metzler inequalities is established.

The main contributions are summarized as follows:

- A consensus-orthogonal decomposition is used to remove the redundant consensus direction and transform the original consensus problem into stabilization of a reduced-order switched disagreement system.
- Unlike conventional results requiring each topology to be strongly connected or to contain a directed spanning tree, only strong connectivity of the union graph is assumed. This condition excludes common nonzero disagreement directions and guarantees that every strictly positive convex combination of the reduced subsystem matrices is Hurwitz.
- A state-dependent switching law is developed through finite-dwell-time Lyapunov–Metzler inequalities. It enforces a prescribed minimum dwell time and stabilizes the disagreement system without requiring the individual subsystem matrices to be Hurwitz.
- A constructive feasibility analysis is established using a lifted representation of the coupled Lyapunov operator. A Metzler matrix is constructed from the Hurwitz convex combination, and a spectral analysis together with a perturbation argument yields an explicit sufficient upper bound on the dwell time.

The remainder of the paper is organized as follows. Section II presents the graph preliminaries, system model, and reduced-order disagreement dynamics. Section III develops the state-dependent dwell-time switching law for the reduced linear switched system and proves its stability under feasible Lyapunov–Metzler inequalities. Section IV proves the existence of feasible Lyapunov–Metzler matrices from strong union connectivity and derives an explicit dwell-time bound. Section V presents numerical simulations. Section VI concludes the paper.

Notations: $\mathbb{R}^n$ denotes the n-dimensional Euclidean space, and $\mathbb{S}^n$ denotes the space of real symmetric $n \times n$ matrices. The symbols $\mathbf{1}_N$ and I_N denote the all-ones vector and identity matrix, respectively. For a symmetric matrix P, $P \succ 0$ and $P \prec 0$ denote positive and negative definiteness. The symbol $\otimes$ denotes the Kronecker product. For square matrices A and B, $A \oplus B = A \otimes I + I \otimes B$ denotes the Kronecker sum, with identity dimensions inferred from context. For vectors, $\| \cdot \|_2$ is the Euclidean norm; for matrices, it is the induced spectral norm. For a square matrix A, $\operatorname{spec}(A)$ denotes its spectrum. The spectral abscissa is defined as $\alpha(A) = \max_{\mu \in \operatorname{spec}(A)} \operatorname{Re}(\mu)$. The notation $\operatorname{col}\{\cdot\}$ denotes column stacking. For a matrix $X \in \mathbb{R}^{m \times n}$, $\operatorname{vec}(X) \in \mathbb{R}^{mn}$ denotes the column-wise vectorization obtained by stacking the columns of X. The class $\mathcal{M}$ contains Metzler matrices $\Pi = [\pi_{ij}] \in \mathbb{R}^{M \times M}$ satisfying $\pi_{ij} \geq 0$ for $i \neq j$ and $\Pi \mathbf{1}_M = 0$.

II. Preliminaries and Problem Formulation

Consider N agents with states $x_i(t) \in \mathbb{R}^d$. Define $x(t) = \operatorname{col}\{x_1(t), \ldots, x_N(t)\} \in \mathbb{R}^{Nd}$. The communication network switches among M candidate directed graphs $\mathcal{G}_p$. The adjacency weight $a_{ij}^p \geq 0$ represents the influence of agent j on agent i, and $[L_p]_{ii} = \sum_{j \neq i} a_{ij}^p$, $[L_p]_{ij} = -a_{ij}^p$, $i \neq j$. Hence, $L_p \mathbf{1}_N = 0$, $p = 1, 2, \ldots, M$. A directed path from node i to node j is a sequence of directed edges leading from i to j. A directed graph is strongly connected if every node is reachable from every other node through a directed path.

Assumption 1. Each candidate directed topology $\mathcal{G}_p$ is disconnected, whereas the union graph

$$\mathcal{G}_U = \bigcup_{p=1}^{M} \mathcal{G}_p$$

is strongly connected.

Under the active topology $\mathcal{G}_{\sigma(t)}$, where $\sigma(t) \in \Omega = \{1, \ldots, M\}$ is the switching signal, the agent dynamics under the standard consensus protocol are given by

$$\dot{x}_i(t) = u_i(t) = -\sum_{j=1}^{N} a_{ij}^{\sigma(t)} \big(x_i(t) - x_j(t) \big), i = 1, \ldots, N. \tag{1}$$

By stacking $x(t) = \operatorname{col}\{x_1(t), \ldots, x_N(t)\}$, then (1) can be written in campact form as

$$\dot{x}(t) = -\big(L_{\sigma(t)} \otimes I_d \big) x(t), \tag{2}$$

where I_d is the d-dimensional identity matrix.

The multi-agent system consensus is achieved when $\lim_{t \to \infty} (x_i(t) - x_j(t)) = 0$ for all $i, j \in \{1, 2, \ldots, N\}$, or equivalently when $x(t)$ converges to the consensus subspace $\mathcal{S}_c = \{\mathbf{1}_N \otimes \eta : \eta \in \mathbb{R}^d\}$.

To describe the disagreement among agents, define the centering matrix $H = I_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top$. Choose $S \in \mathbb{R}^{(N-1) \times N}$ such that $S \mathbf{1}_N = 0$, $S S^\top = I_{N-1}$, $S^\top S = H$. Thus, the rows of S form an orthonormal basis of the subspace orthogonal to the consensus direction $\mathbf{1}_N$.

Define the reduced-order disagreement coordinate $\xi(t) = (S \otimes I_d) x(t) \in \mathbb{R}^r$, $r = (N-1)d$. Since $\operatorname{rank}(S) = N - 1$ and $S \mathbf{1}_N = 0$, we have $\ker(S) = \operatorname{span}\{\mathbf{1}_N\}$, which implies $\xi(t) = 0$ if and only if there exists $\eta(t) \in \mathbb{R}^d$ such that $x(t) = \mathbf{1}_N \otimes \eta(t)$. Hence, $\xi(t) \to 0$ is equivalent to consensus of the original multi-agent system.

Define the arithmetic-average coordinate $\eta(t) = \frac{1}{N}(\mathbf{1}_N^\top \otimes I_d)x(t)$. Using $I_N = \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top + S^\top S$, the original state can be decomposed as

$$x(t) = \mathbf{1}_N \otimes \eta(t) + (S^\top \otimes I_d)\xi(t). \tag{3}$$

Differentiating (II) and using (2) and (3) gives

$$\begin{aligned}\dot{\xi}(t) &= -(SL_{\sigma(t)} \otimes I_d)x(t) \\ &= -\big(SL_{\sigma(t)}\mathbf{1}_N \otimes I_d\big)\eta(t) - \big(SL_{\sigma(t)}S^\top \otimes I_d\big)\xi(t) \\ &= -\big(SL_{\sigma(t)}S^\top \otimes I_d\big)\xi(t).\end{aligned} \tag{4}$$

Define $\bar{A}_p = -\big(SL_pS^\top \otimes I_d\big)$. Then, the reduced-order disagreement dynamics are represented by the switched linear system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t). \tag{5}$$

III. State-Dependent Dwell-Time Switching Design for the Reduced Linear System

Consider the reduced-order disagreement error system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t).$$

Define the disagreement output as $z(t) = C\xi(t)$, where C is a given full-column-rank matrix. In particular, $C = I_r$ gives direct measurement of the full disagreement state.

For each mode $i \in \Omega$, choose a positive definite matrix $X_i = X_i^\top > 0$ and define the Lyapunov function as $V_i(\xi) = \xi^\top X_i \xi$. Let $\Pi = [\pi_{ij}] \in \mathcal{M}$ be a Metzler matrix satisfying $\pi_{ij} \geq 0$ for $i \neq j$ and $\sum_{j=1}^{M} \pi_{ij} = 0$. Given a prescribed minimum dwell time $T > 0$, define, for each mode j,

$$Y_{1,j} = e^{\bar{A}_j^\top T} X_j e^{\bar{A}_j T}, \quad Y_{2,j} = \int_0^T e^{\bar{A}_j^\top \tau} C^\top C e^{\bar{A}_j \tau} d\tau.$$

The following finite-dwell-time Lyapunov–Metzler inequalities are introduced as the main stability condition:

$$\begin{aligned}&\bar{A}_i^\top X_i + X_i \bar{A}_i + \sum_{j\neq i} \pi_{ij}\left(Y_{1,j} + Y_{2,j} - X_i\right) + C^\top C < 0, \\ &i = 1, 2, \ldots, M.\end{aligned} \tag{6}$$

For a given active mode $i = \sigma(t_k)$, define $\Omega_i = \{1, 2, \ldots, M\} \setminus \{i\}$. For $j \in \Omega_i$, define $\Delta_{ij}(t) = \xi^\top(t)\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t)$ and $J_j(t) = \xi^\top(t)\left(Y_{1,j} + Y_{2,j}\right)\xi(t)$. Then, the proposed state-dependent dwell-time switching law is given by

$$\begin{aligned}&\sigma(t) = i, && t \in [t_k, t_k + T], \\ &\sigma(t) = i, && t > t_k + T, \\ & && \Delta_{ij}(t) \geq 0,\ \forall j \in \Omega_i, \\ &\sigma(t_{k+1}) = \arg\min_{j\in\Omega_i} J_j(t_{k+1}), && \text{otherwise.}\end{aligned} \tag{7}$$

where $t_{k+1} = \inf_{t>t_k+T}\left\{t : \exists j \in \Omega_i,\ \Delta_{ij}(t) < 0\right\}$.

Theorem 1. Consider the reduced-order disagreement error system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t).$$

If there exist positive definite matrices $X_i = X_i^\top > 0$, $i = 1, 2, \ldots, M$, and a Metzler matrix Π such that the Lyapunov–Metzler inequalities

$$\bar{A}_i^\top X_i + X_i \bar{A}_i + \sum_{j\neq i} \pi_{ij}\left(Y_{1,j} + Y_{2,j} - X_i\right) + C^\top C < 0.$$

Then under the proposed state-dependent dwell-time switching law, the disagreement error system is asymptotically stable. In particular, $\xi(t) \to 0$ as $t \to \infty$, which implies global consensus of the original multi-agent system:

$$\lim_{t\to\infty}\big(x_i(t) - x_j(t)\big) = 0, \qquad \forall i, j \in \{1, \ldots, N\}.$$

Proof. Let t_k be a switching instant and suppose that the active mode is i. Define the following piecewise Lyapunov function:

$$\begin{aligned}V(\xi, t) &= \xi^\top(t)P_i(t)\xi(t), \quad t \in [t_k, t_k + T), \\ V(\xi, t) &= \xi^\top(t)X_i\xi(t), \quad t \in [t_k + T, t_{k+1}).\end{aligned} \tag{8}$$

Here, $P_i(t)$ satisfies

$$-\dot{P}_i(t) = \bar{A}_i^\top P_i(t) + P_i(t)\bar{A}_i + C^\top C, \tag{9}$$

with the terminal condition $P_i(t_k + T) = X_i$.

On the forced dwell-time interval $[t_k, t_k + T)$, one has $\sigma(t) = i$. Using the error dynamics and the above differential Lyapunov equation gives

$$\begin{aligned}\dot{V}(\xi, t) &= \xi^\top(t)\left(\bar{A}_i^\top P_i(t) + P_i(t)\bar{A}_i + \dot{P}_i(t)\right)\xi(t) \\ &= -\xi^\top(t)C^\top C\xi(t).\end{aligned} \tag{10}$$

Thus, the Lyapunov function is non-increasing during the forced dwell-time interval.

For $t \in [t_k + T, t_{k+1})$, no switch has been triggered. Hence,

$$\xi^\top(t)\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t) \geq 0, \quad \forall j \neq i.$$

Since $\pi_{ij} \geq 0$, it follows from the Lyapunov–Metzler inequality that

$$\begin{aligned}\dot{V}(\xi, t) &= \xi^\top(t)\left(\bar{A}_i^\top X_i + X_i\bar{A}_i\right)\xi(t) \\ &< -\xi^\top(t)\left[\sum_{j\neq i}\pi_{ij}\left(Y_{1,j} + Y_{2,j} - X_i\right) + C^\top C\right]\xi(t) \\ &= -\xi^\top(t)\left[\sum_{j\neq i}\pi_{ij}\left(Y_{1,j} + Y_{2,j} - X_i\right)\right]\xi(t) \\ &\quad - \xi^\top(t)C^\top C\xi(t) \\ &\leq -\xi^\top(t)C^\top C\xi(t).\end{aligned} \tag{11}$$

Therefore, the Lyapunov function remains non-increasing before the next switching instant.

At the switching instant t_{k+1}, the switching condition ensures that there exists some $j \neq i$ satisfying

$$\xi^\top(t_{k+1})\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t_{k+1}) < 0. \tag{12}$$

Moreover, the new mode is selected by minimizing the predicted dwell-time cost. Therefore,

$$\begin{aligned}\Delta V(t_{k+1}) &= V(\xi(t_{k+1}), t_{k+1}) - V(\xi(t_{k+1}^-), t_{k+1}^-)\\ &\le \xi^\top(t_{k+1})\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t_{k+1})\\ &< 0.\end{aligned} \tag{13}$$

Thus, each switching instant yields a strict decrease of the Lyapunov function.

Combining the continuous evolution and the switching instants yields

$$\dot V(\xi, t) \le -\xi^\top(t) C^\top C \xi(t), \qquad \Delta V(t_k) < 0.$$

By the positive definiteness of V and standard stability arguments, $\xi(t) \to 0$, which is equivalent to global consensus of the original multi-agent system. □

Remark 1. The Lyapunov–Metzler inequalities in (6) do not require any individual subsystem matrix $\bar A_i$ to be Hurwitz. In particular, the candidate topologies may be disconnected and the corresponding reduced-order subsystems may contain non-decaying disagreement modes. Consensus is instead achieved through the proposed state-dependent dwell-time switching among these subsystems.

IV. Existence and Design of Metzler matrix

This section proves that under Assumption 1, there exists a feasible solution to the finite-dwell-time Lyapunov–Metzler inequalities required by Theorem 1.

Choose any vector $\lambda = [\lambda_1, \ldots, \lambda_M]^\top \in \mathbb{R}^M$ satisfying $\lambda_p > 0$, $\sum_{p=1}^M \lambda_p = 1$, $p = 1, \ldots, M$. Define $L_\lambda = \sum_{p=1}^M \lambda_p L_p$, Then

$$\bar A_\lambda = \sum_{p=1}^M \lambda_p \bar A_p = -\big(S L_\lambda S^\top \otimes I_d\big).$$

Lemma 1. Under Assumption 1, $\bar A_\lambda$ is Hurwitz for every strictly positive convex weight vector λ.

Proof. Because all λ_p are strictly positive and all adjacency weights are nonnegative, an edge has positive weight in the graph associated with L_λ if and only if it appears in at least one candidate graph. Hence this weighted graph has the same edge set as the union graph and is strongly connected.

Define $\mathcal{T} = \begin{bmatrix}\mathbf{1}_N & S^\top\end{bmatrix}$, $\mathcal{T}^{-1} = \begin{bmatrix}\frac{1}{N}\mathbf{1}_N^\top \\ S\end{bmatrix}$. Using $L_\lambda \mathbf{1}_N = 0$ gives $\mathcal{T}^{-1} L_\lambda \mathcal{T} = \begin{bmatrix}0 & \frac{1}{N}\mathbf{1}_N^\top L_\lambda S^\top \\ 0 & S L_\lambda S^\top\end{bmatrix}$. Therefore,

$$\operatorname{spec}(L_\lambda) = \operatorname{spec}(\mathcal{T}^{-1} L_\lambda \mathcal{T}) = \{0\} \cup \operatorname{spec}(S L_\lambda S^\top).$$

Strong connectivity implies that the zero eigenvalue of L_λ is simple and all remaining eigenvalues have strictly positive real parts. Hence every eigenvalue of $S L_\lambda S^\top$ has strictly positive real part. Consequently, every eigenvalue of $\bar A_\lambda$ has strictly negative real part, so $\bar A_\lambda$ is Hurwitz. □

Define $E_i(T) = e^{\bar A_i T}$ and $\mathcal{B}_i = \bar A_i^\top \oplus \bar A_i^\top = \bar A_i^\top \otimes I_r + I_r \otimes \bar A_i^\top \in \mathbb{R}^{r^2 \times r^2}$. Furthermore, let $\mathscr{B} = \operatorname{diag}\{\mathcal{B}_1, \ldots, \mathcal{B}_M\}, \Pi_d = \operatorname{diag}\{\pi_{11}, \ldots, \pi_{MM}\}$. For $\mathbf{X} = (X_1, \ldots, X_M)$, define the homogeneous linear operator

$$\begin{aligned}[\mathfrak{F}_T(\mathbf{X})]_i = \bar A_i^\top X_i + X_i \bar A_i + \pi_{ii} X_i\\ + \sum_{j \ne i} \pi_{ij} E_j^\top(T) X_j E_j(T).\end{aligned} \tag{14}$$

Using $\operatorname{vec}(AXB) = (B^\top \otimes A)\operatorname{vec}(X)$ and $E_j^\top(T) \otimes E_j^\top(T) = e^{\mathcal{B}_j T}$, then

$$\begin{aligned}&\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{X})]_i\right)\\ &= \left[I_r \otimes \bar A_i^\top + \bar A_i^\top \otimes I_r + \pi_{ii} I_{r^2}\right]\operatorname{vec}(X_i)\\ &\quad + \sum_{j \ne i} \pi_{ij}\left(E_j^\top(T) \otimes E_j^\top(T)\right)\operatorname{vec}(X_j)\\ &= \left(\mathcal{B}_i + \pi_{ii} I_{r^2}\right)\operatorname{vec}(X_i) + \sum_{j \ne i} \pi_{ij} e^{\mathcal{B}_j T}\operatorname{vec}(X_j).\end{aligned} \tag{15}$$

Stacking (15) for $i = 1, \ldots, M$ yields

$$\operatorname{col}\left\{\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{X})]_i\right)\right\}_{i=1}^M = \mathcal{J}_T \operatorname{col}\left\{\operatorname{vec}(X_i)\right\}_{i=1}^M, \tag{16}$$

where the (i, j)th block of $\mathcal{J}_T$ is

$$[\mathcal{J}_T]_{ij} = \begin{cases}\mathcal{B}_i + \pi_{ii} I_{r^2}, & i = j,\\ \pi_{ij} e^{\mathcal{B}_j T}, & i \ne j.\end{cases}$$

Since $e^{\mathscr{B} T} = \operatorname{diag}\left\{e^{\mathcal{B}_1 T}, \ldots, e^{\mathcal{B}_M T}\right\}$, the above block matrix can be written compactly as

$$\mathcal{J}_T = \mathscr{B} + \Pi_d \otimes I_{r^2} + \left[(\Pi - \Pi_d) \otimes I_{r^2}\right] e^{\mathscr{B} T}. \tag{17}$$

Lemma 2. Given a Metzler matrix $\Pi \in \mathcal{M}$, a prescribed dwell time $T > 0$, and a matrix C satisfying $Q = C^\top C \succ 0$, define $\mathcal{J}_T$ by (17). If $\mathcal{J}_T$ is Hurwitz, then the Lyapunov–Metzler inequalities (6) admit symmetric positive-definite solutions

$$X_i = X_i^\top \succ 0, \qquad i = 1, \ldots, M.$$

Proof. Define $W_i(T) = Q + \sum_{j \ne i} \pi_{ij} Y_{2,j}(T)$. Since $Q \succ 0$, $\pi_{ij} \ge 0$ for $i \ne j$, then $W_i(T) \succ 0$. The Lyapunov–Metzler inequalities can be written as

$$[\mathfrak{F}_T(\mathbf{X})]_i + W_i(T) \prec 0. \tag{18}$$

Suppose that $\mathcal{J}_T$ is Hurwitz, Consider the auxiliary matrix-valued linear system

$$\dot{\mathbf{Z}}(s) = \mathfrak{F}_T\big(\mathbf{Z}(s)\big), \qquad \mathbf{Z}(s) = \big(Z_1(s), \ldots, Z_M(s)\big), \tag{19}$$

where $s \ge 0$ is an auxiliary variable used in the construction of the Lyapunov matrices.

By the stacked vectorization representation derived above,

$$\operatorname{col}\left\{\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{Z})]_i\right)\right\}_{i=1}^M = \mathcal{J}_T \operatorname{col}\left\{\operatorname{vec}(Z_i)\right\}_{i=1}^M.$$

Define

$$\mathbf{z}(s) = \operatorname{col}\left\{\operatorname{vec}(Z_i(s))\right\}_{i=1}^M.$$

Then (19) is equivalent to the finite-dimensional linear system

$$\dot{\mathbf{z}}(s) = \mathcal{J}_T \mathbf{z}(s).$$

Since $\mathcal{J}_T$ is Hurwitz, there exist constants $c > 0$ and $\gamma > 0$ such that

$$\left\|e^{\mathcal{J}_T s}\right\|_2 \leq c e^{-\gamma s}, \qquad s \geq 0.$$

Consequently,

$$\left\|e^{\mathfrak{F}_T s}\mathbf{Z}(0)\right\| \leq c e^{-\gamma s} \left\|\mathbf{Z}(0)\right\|, \qquad s \geq 0.$$

Hence, the operator semigroup $e^{\mathfrak{F}_T s}$ is exponentially stable.

We next show that the semigroup $e^{\mathfrak{F}_T s}$ preserves the positive-semidefinite cone. Define

$$\widetilde{A}_i = \bar{A}_i + \frac{\pi_{ii}}{2} I_r.$$

Then the ith component of (19) can be written as

$$\begin{aligned} \dot{Z}_i &= \widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i \\ &\quad + \sum_{j\neq i} \pi_{ij} E_j^\top(T) Z_j E_j(T). \end{aligned} \tag{20}$$

Indeed,

$$\begin{aligned} \widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i &= \left(\bar{A}_i^\top + \frac{\pi_{ii}}{2} I_r\right) Z_i + Z_i \left(\bar{A}_i + \frac{\pi_{ii}}{2} I_r\right) \\ &= \bar{A}_i^\top Z_i + Z_i \bar{A}_i + \pi_{ii} Z_i. \end{aligned}$$

In the absence of the coupling terms, the corresponding matrix dynamics is

$$\dot{Z}_i = \widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i,$$

whose solution is

$$Z_i(s) = e^{\widetilde{A}_i^\top s} Z_i(0) e^{\widetilde{A}_i s}.$$

Therefore, if $Z_i(0) \succeq 0$, then

$$e^{\widetilde{A}_i^\top s} Z_i(0) e^{\widetilde{A}_i s} \succeq 0,$$

because congruence transformations preserve positive semidefiniteness.

Furthermore, since Π is Metzler,

$$\pi_{ij} \geq 0, \qquad i \neq j.$$

Thus, whenever $Z_j \succeq 0$, one has

$$\pi_{ij} E_j^\top(T) Z_j E_j(T) \succeq 0.$$

Hence, each off-diagonal coupling term in (20) is positive semidefinite.

For a rigorous cone-invariance argument, consider a point on the boundary of $(\mathbb{S}_+^r)^M$. Suppose that $Z_i \succeq 0$ and let $v \in \ker(Z_i)$. Then

$$Z_i v = 0.$$

Premultiplying and postmultiplying (20) by $v^\top$ and v, respectively, gives

$$\begin{aligned} v^\top \dot{Z}_i v &= v^\top \left(\widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i\right) v \\ &\quad + \sum_{j\neq i} \pi_{ij} v^\top E_j^\top(T) Z_j E_j(T) v. \end{aligned} \tag{21}$$

Since $Z_i v = 0$, the first term in (21) vanishes:

$$v^\top \left(\widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i\right) v = 0.$$

Therefore,

$$\begin{aligned} v^\top \dot{Z}_i v &= \sum_{j\neq i} \pi_{ij} v^\top E_j^\top(T) Z_j E_j(T) v \\ &= \sum_{j\neq i} \pi_{ij} \big(E_j(T) v\big)^\top Z_j \big(E_j(T) v\big) \\ &\geq 0. \end{aligned} \tag{22}$$

Equation (22) shows that, at every boundary point of the positive-semidefinite cone, the vector field cannot point outside the cone. Consequently,

$$\mathbf{Z}(0) \in (\mathbb{S}_+^r)^M \quad \Longrightarrow \quad e^{\mathfrak{F}_T s}\mathbf{Z}(0) \in (\mathbb{S}_+^r)^M, \qquad s \geq 0.$$

Thus, $e^{\mathfrak{F}_T s}$ is a positive semigroup on $(\mathbb{S}_+^r)^M$. Here, positivity means preservation of the positive-semidefinite cone, rather than entrywise nonnegativity.

Choose an arbitrary scalar $\rho > 1$ and define

$$R_i = \rho W_i(T) \succ 0, \qquad i = 1, \ldots, M.$$

Let

$$\mathbf{R} = (R_1, \ldots, R_M)$$

and construct

$$\mathbf{X} = \int_0^\infty e^{\mathfrak{F}_T s}\mathbf{R}\, ds. \tag{23}$$

Since $e^{\mathfrak{F}_T s}$ is exponentially stable, there exist $c > 0$ and $\gamma > 0$ such that

$$\left\|e^{\mathfrak{F}_T s}\mathbf{R}\right\| \leq c e^{-\gamma s} \|\mathbf{R}\|.$$

Hence, the integral in (23) converges.

Moreover, $\mathfrak{F}_T$ maps symmetric matrix tuples into symmetric matrix tuples. Indeed, if every Z_i is symmetric, then

$$\left(\bar{A}_i^\top Z_i + Z_i \bar{A}_i\right)^\top = \bar{A}_i^\top Z_i + Z_i \bar{A}_i,$$

and

$$\left(E_j^\top(T) Z_j E_j(T)\right)^\top = E_j^\top(T) Z_j E_j(T).$$

Since every R_i is symmetric, each component of $e^{\mathfrak{F}_T s}\mathbf{R}$ is symmetric. It follows that

$$X_i = X_i^\top, \qquad i = 1, \ldots, M.$$

The positivity of the semigroup further gives

$$\left[e^{\mathfrak{F}_T s}\mathbf{R}\right]_i \succeq 0, \qquad s \geq 0.$$

Consequently,

$$X_i = \int_0^\infty \left[e^{\mathfrak{F}_T s}\mathbf{R}\right]_i\, ds \succeq 0.$$

To establish strict positive definiteness, observe that

$$\left[e^{\mathfrak{F}_T 0}\mathbf{R}\right]_i = R_i = \rho W_i(T) \succ 0.$$

Since the mapping

$$s \longmapsto \left[e^{\mathfrak{F}_T s}\mathbf{R}\right]_i$$

is continuous, there exists $\delta_i > 0$ such that

$$\left[e^{\mathfrak{F}_T s}\mathbf{R}\right]_i \succ 0, \qquad 0 \leq s \leq \delta_i.$$

Therefore,

$$\begin{aligned} X_i &= \int_0^\infty \left[e^{\mathfrak{F}_T s}\mathbf{R}\right]_i \, ds \\ &\succeq \int_0^{\delta_i} \left[e^{\mathfrak{F}_T s}\mathbf{R}\right]_i \, ds \\ &\succ 0. \end{aligned}$$

Thus,

$$X_i = X_i^\top \succ 0, \qquad i = 1, \ldots, M.$$

Finally, using the exponential stability of the semigroup and the linearity of $\mathfrak{F}_T$, one obtains

$$\begin{aligned} \mathfrak{F}_T(\mathbf{X}) &= \mathfrak{F}_T\left(\int_0^\infty e^{\mathfrak{F}_T s}\mathbf{R}\, ds\right) \\ &= \int_0^\infty \mathfrak{F}_T e^{\mathfrak{F}_T s}\mathbf{R}\, ds \\ &= \int_0^\infty \frac{d}{ds}\left(e^{\mathfrak{F}_T s}\mathbf{R}\right) ds \\ &= \lim_{s\to\infty} e^{\mathfrak{F}_T s}\mathbf{R} - e^{\mathfrak{F}_T 0}\mathbf{R} \\ &= -\mathbf{R}. \end{aligned}$$

Therefore, for every $i = 1, \ldots, M$,

$$\begin{aligned} [\mathfrak{F}_T(\mathbf{X})]_i + W_i(T) &= -R_i + W_i(T) \\ &= -\rho W_i(T) + W_i(T) \\ &= -(\rho - 1) W_i(T) \\ &\prec 0, \end{aligned}$$

where the last inequality follows from

$$\rho > 1, \qquad W_i(T) \succ 0.$$

Hence, the Lyapunov–Metzler inequalities (6) admit symmetric positive-definite solutions. $\square$

Define

$$\Pi_0 = \mathbf{1}_M \lambda^\top - I_M, \qquad \Pi_\kappa = \kappa \Pi_0, \qquad \kappa > 0.$$

Since every entry of λ is strictly positive, Π_κ is a Metzler matrix. Moreover,

$$\Pi_\kappa \mathbf{1}_M = \kappa\left(\mathbf{1}_M \lambda^\top \mathbf{1}_M - \mathbf{1}_M\right) = \mathbf{0},$$

because $\lambda^\top \mathbf{1}_M = 1$, and

$$\lambda^\top \Pi_\kappa = \kappa\left(\lambda^\top \mathbf{1}_M \lambda^\top - \lambda^\top\right) = \mathbf{0}.$$

At $T = 0$, one has

$$e^{\mathscr{B}T} = I_{Mr^2},$$

and therefore (17) reduces to

$$\mathcal{J}_0(\Pi_\kappa) = \mathscr{B} + \Pi_\kappa \otimes I_{r^2}. \tag{24}$$

Lemma 3. Under Assumption 1, there exists $\kappa^\star > 0$ such that $\mathcal{J}_0(\Pi_\kappa)$ is Hurwitz for every $\kappa > \kappa^\star$.

Proof. The matrix

$$\Pi_0 = \mathbf{1}_M \lambda^\top - I_M$$

has a simple zero eigenvalue with right eigenvector $\mathbf{1}_M$ and left eigenvector λ. Indeed,

$$\Pi_0 \mathbf{1}_M = \mathbf{0}, \qquad \lambda^\top \Pi_0 = \mathbf{0}.$$

Moreover, all its remaining $M - 1$ eigenvalues are equal to -1.

Choose a matrix

$$\Phi \in \mathbb{R}^{M\times(M-1)}$$

such that

$$\lambda^\top \Phi = \mathbf{0}$$

and

$$U_M = [\mathbf{1}_M, \Phi]$$

is nonsingular. Write

$$U_M^{-1} = \begin{bmatrix} \lambda^\top \\ W^\top \end{bmatrix}.$$

Then

$$U_M^{-1} \Pi_0 U_M = \operatorname{diag}\{0, -I_{M-1}\}.$$

Define

$$U = U_M \otimes I_{r^2}.$$

A similarity transformation gives

$$U^{-1}\mathcal{J}_0(\Pi_\kappa)U = \begin{bmatrix} \bar{\mathcal{B}} & \mathcal{B}_{12} \\ \mathcal{B}_{21} & \mathcal{B}_{22} - \kappa I_{(M-1)r^2} \end{bmatrix}, \tag{25}$$

where

$$\bar{\mathcal{B}} = \sum_{i=1}^M \lambda_i \mathcal{B}_i.$$

Using

$$\mathcal{B}_i = \bar{A}_i^\top \oplus \bar{A}_i^\top,$$

one obtains

$$\begin{aligned} \bar{\mathcal{B}} &= \sum_{i=1}^M \lambda_i \left(\bar{A}_i^\top \oplus \bar{A}_i^\top\right) \\ &= \left(\sum_{i=1}^M \lambda_i \bar{A}_i^\top\right) \oplus \left(\sum_{i=1}^M \lambda_i \bar{A}_i^\top\right) \\ &= \bar{A}_\lambda^\top \oplus \bar{A}_\lambda^\top. \end{aligned}$$

By Lemma 1, $\bar{A}_\lambda$ is Hurwitz. Therefore, $\bar{A}_\lambda^\top$ is also Hurwitz.

The eigenvalues of the Kronecker sum

$$\bar{A}_\lambda^\top \oplus \bar{A}_\lambda^\top$$

are all pairwise sums

$$\mu_p + \mu_q, \qquad \mu_p, \mu_q \in \operatorname{spec}(\bar{A}_\lambda^\top).$$

Since

$$\operatorname{Re}(\mu_p) < 0, \qquad \operatorname{Re}(\mu_q) < 0,$$

it follows that

$$\operatorname{Re}(\mu_p + \mu_q) < 0.$$

Hence, $\bar{\mathcal{B}}$ is Hurwitz.

As $\kappa \to \infty$, the spectrum of the transformed matrix (25) separates into two groups. Consider first the eigenvalues $s = s(\kappa)$ that remain bounded. Applying the Schur-complement formula gives

$$\det\left[sI_{r^2} - \bar{\mathcal{B}} - \mathcal{B}_{12}\left(sI_{(M-1)r^2} - \mathcal{B}_{22} + \kappa I_{(M-1)r^2}\right)^{-1}\mathcal{B}_{21}\right] = 0.$$

Since

$$\left(sI_{(M-1)r^2} - \mathcal{B}_{22} + \kappa I_{(M-1)r^2}\right)^{-1} = O(\kappa^{-1}),$$

there are r^2 eigenvalues converging to

$$\mathrm{spec}(\bar{\mathcal{B}}).$$

Since $\bar{\mathcal{B}}$ is Hurwitz, these eigenvalues lie in the open left half-plane for all sufficiently large κ.

For the remaining $(M-1)r^2$ eigenvalues, set

$$s = -\kappa + \mu.$$

A second Schur-complement argument gives

$$\det\left[\mu I_{(M-1)r^2} - \mathcal{B}_{22} - \mathcal{B}_{21}\left((-\kappa+\mu)I_{r^2} - \bar{\mathcal{B}}\right)^{-1}\mathcal{B}_{12}\right] = 0.$$

Since

$$\left((-\kappa+\mu)I_{r^2} - \bar{\mathcal{B}}\right)^{-1} = O(\kappa^{-1}),$$

one has

$$\mu = O(1),$$

and therefore

$$s = -\kappa + O(1).$$

Thus, these $(M-1)r^2$ eigenvalues move to the open left half-plane as κ increases.

Consequently, the first r^2 eigenvalues approach the spectrum of the Hurwitz matrix $\bar{\mathcal{B}}$, whereas the remaining $(M-1)r^2$ eigenvalues move toward minus infinity. Hence, there exists $\kappa^\star > 0$ such that

$$\mathcal{J}_0(\Pi_\kappa)$$

is Hurwitz for every $\kappa > \kappa^\star$. □

Fix a Metzler matrix Π obtained from Lemma 3 such that $\mathcal{J}_0(\Pi)$ is Hurwitz. Let $P = P^\top \succ 0$ be the unique solution of

$$\mathcal{J}_0^\top P + P\mathcal{J}_0 = -I_{Mr^2}. \tag{26}$$

Define

$$\Delta_T = \mathcal{J}_T - \mathcal{J}_0.$$

From (17) and (24), one obtains

$$\begin{aligned}\Delta_T &= \mathscr{B} + \Pi_d \otimes I_{r^2} + \left[(\Pi - \Pi_d)\otimes I_{r^2}\right]e^{\mathscr{B}T} \\ &\quad - \mathscr{B} - \Pi \otimes I_{r^2} \\ &= \left[(\Pi - \Pi_d)\otimes I_{r^2}\right]\left(e^{\mathscr{B}T} - I_{Mr^2}\right).\end{aligned} \tag{27}$$

Theorem 2. Suppose that Assumption 1 holds and

$$Q = C^\top C \succ 0.$$

Choose a strictly positive convex vector

$$\lambda = [\lambda_1, \dots, \lambda_M]^\top, \qquad \lambda_i > 0, \qquad \sum_{i=1}^{M}\lambda_i = 1,$$

and choose a sufficiently large $\kappa > 0$ such that

$$\Pi = \kappa\left(\mathbf{1}_M\lambda^\top - I_M\right)$$

makes $\mathcal{J}_0(\Pi)$ Hurwitz. Let $P = P^\top \succ 0$ be the solution of (26), and define

$$T_{\mathrm{ub}} = \frac{1}{\|\mathscr{B}\|_2}\ln\left(1 + \frac{1}{2\|P\|_2\|\Pi - \Pi_d\|_2}\right). \tag{28}$$

Then, for every prescribed dwell time satisfying

$$0 < T < T_{\mathrm{ub}},$$

the finite-dwell-time Lyapunov–Metzler inequalities (6) admit symmetric positive-definite solutions

$$X_i = X_i^\top \succ 0, \qquad i = 1, \dots, M.$$

Proof. Using (27) and the submultiplicativity of the spectral norm gives

$$\begin{aligned}\|\Delta_T\|_2 &\le \|(\Pi - \Pi_d)\otimes I_{r^2}\|_2\left\|e^{\mathscr{B}T} - I_{Mr^2}\right\|_2 \\ &= \|\Pi - \Pi_d\|_2\left\|e^{\mathscr{B}T} - I_{Mr^2}\right\|_2.\end{aligned} \tag{29}$$

Using

$$e^{\mathscr{B}T} - I_{Mr^2} = \int_0^T \mathscr{B}e^{\mathscr{B}s}\,ds,$$

one obtains

$$\begin{aligned}\left\|e^{\mathscr{B}T} - I_{Mr^2}\right\|_2 &\le \int_0^T \|\mathscr{B}\|_2\left\|e^{\mathscr{B}s}\right\|_2\,ds \\ &\le \int_0^T \|\mathscr{B}\|_2 e^{\|\mathscr{B}\|_2 s}\,ds \\ &= e^{\|\mathscr{B}\|_2 T} - 1.\end{aligned} \tag{30}$$

Combining (29) and (30) gives

$$\|\Delta_T\|_2 \le \|\Pi - \Pi_d\|_2\left(e^{\|\mathscr{B}\|_2 T} - 1\right). \tag{31}$$

Since

$$\mathcal{J}_T = \mathcal{J}_0 + \Delta_T,$$

it follows from (26) that

$$\begin{aligned}\mathcal{J}_T^\top P + P\mathcal{J}_T &= \mathcal{J}_0^\top P + P\mathcal{J}_0 + \Delta_T^\top P + P\Delta_T \\ &= -I_{Mr^2} + \Delta_T^\top P + P\Delta_T.\end{aligned} \tag{32}$$

For every vector $y \in \mathbb{R}^{Mr^2}$,

$$\begin{aligned}y^\top\left(\Delta_T^\top P + P\Delta_T\right)y &= 2y^\top P\Delta_T y \\ &\le 2\|P\|_2\|\Delta_T\|_2\|y\|^2.\end{aligned}$$

Therefore,

$$\Delta_T^\top P + P\Delta_T \preceq 2\|P\|_2\|\Delta_T\|_2 I_{Mr^2}.$$

Substituting this inequality into (32) yields

$$\mathcal{J}_T^\top P + P\mathcal{J}_T \preceq -\left[1 - 2\|P\|_2\|\Delta_T\|_2\right]I_{Mr^2}. \tag{33}$$

A sufficient condition for the right-hand side of (33) to be negative definite is

$$2\|P\|_2\|\Delta_T\|_2 < 1.$$

Using (31), this condition is guaranteed by

$$2\|P\|_2\|\Pi - \Pi_d\|_2 \left(e^{\|\mathscr{B}\|_2 T} - 1\right) < 1.$$

Equivalently,

$$e^{\|\mathscr{B}\|_2 T} < 1 + \frac{1}{2\|P\|_2\|\Pi - \Pi_d\|_2},$$

which gives

$$T < \frac{1}{\|\mathscr{B}\|_2} \ln\left(1 + \frac{1}{2\|P\|_2\|\Pi - \Pi_d\|_2}\right) = T_{\text{ub}}.$$

Therefore, for every

$$0 < T < T_{\text{ub}},$$

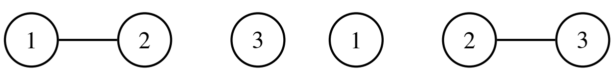
one has

$$\mathscr{J}_T^\top P + P\mathscr{J}_T \prec 0.$$

Hence, $\mathscr{J}_T$ is Hurwitz. By Lemma 2, the Lyapunov–Metzler inequalities (6) admit symmetric positive-definite solutions. □

V. Numerical Simulations

Consider a multi-agent system with $N = 3$, $d = 1$, and two candidate topologies, as shown in Fig. 1. Each topology contains an isolated node, whereas their union graph is strongly connected.



(a) $\mathcal{G}_1$ (b) $\mathcal{G}_2$

Fig. 1. Candidate communication topologies.

Choose $S = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \end{bmatrix}$. The reduced-order system matrices are $\bar{A}_1 = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$, $\bar{A}_2 = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{3}{2} \end{bmatrix}$. Both matrices have eigenvalues $\{-2, 0\}$ and are therefore not Hurwitz. However, for $\lambda = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}^\top$, the convex combination $\bar{A}_\lambda = \frac{1}{2}\bar{A}_1 + \frac{1}{2}\bar{A}_2$ has eigenvalues -1.5 and -0.5.

Set $\Pi = \begin{bmatrix} -0.27 & 0.27 \\ 0.27 & -0.27 \end{bmatrix}$, $C = I_2$.

The analytical dwell-time bound obtained from Theorem 2 is $T_{\text{ub}} = 0.1275$ s. Accordingly, the prescribed dwell time is selected as $T = 0.1$ s $< T_{\text{ub}}$. Therefore, by Theorem 2, the finite-dwell-time Lyapunov–Metzler inequalities are feasible. One feasible solution is

$$X_1 = \begin{bmatrix} 209.77 & 29.40 \\ 29.40 & 954.13 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 742.58 & 337.01 \\ 337.01 & 421.32 \end{bmatrix}.$$

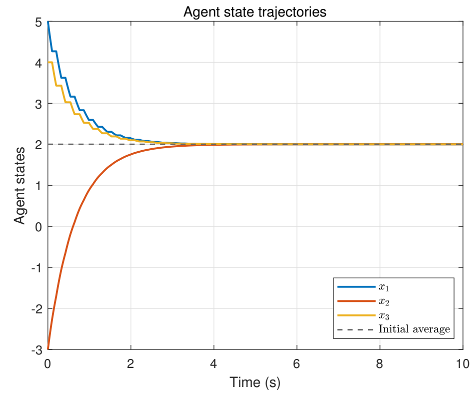


Fig. 2. Agent state trajectories under the proposed switching law.

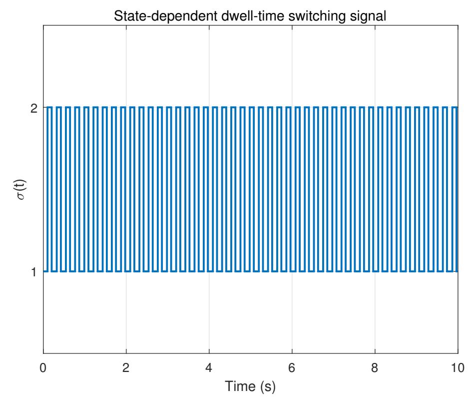


Fig. 3. State-dependent switching signal.

The maximum eigenvalues of the corresponding residual matrices are approximately -156.2373, confirming strict feasibility.

For the initial condition $x(0) = \begin{bmatrix} 5 & -3 & 4 \end{bmatrix}^\top$, $\sigma(0) = 1$, the agent states converge to the common value 2.

As shown in Fig. 2, the three agent states converge from their different initial values to the common value 2. Moreover, the switching signal is shown in Fig. 3. It can be seen that the system switches between the two candidate topologies, and the time interval between any two consecutive switches is no less than the prescribed dwell time T = 0.1. These results confirm that the proposed switching law achieves consensus although neither candidate topology can achieve global consensus independently.

The simulation confirms that consensus can be achieved by switching between individually disconnected topologies, even though neither reduced subsystem matrix is Hurwitz.

VI. Conclusion

This paper investigated the consensus problem for switched multi-agent systems with individually disconnected candidate communication topologies. By introducing a consensus-orthogonal transformation, the original consensus problem was converted into the stabilization of a reduced-order disagreement system. A state-dependent switching law subject to a prescribed minimum dwell time was then developed based on Lyapunov–Metzler inequalities.

Under the assumption that the union graph of all candidate topologies is strongly connected, a Hurwitz convex combination of the reduced-order subsystem matrices was obtained and used to construct a row-sum-zero Metzler matrix. A lifted representation and a perturbation argument were further employed to establish the feasibility of the finite-dwell-time Lyapunov–Metzler inequalities and to derive an explicit sufficient upper bound on the dwell time. The numerical example confirmed that the proposed method achieves global consensus even though none of the individual subsystem matrices is Hurwitz.

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